Gowtham's AI/ML Engineering Roadmap

16 Weeks (112 Days) — Fresher (L1 Support, Fieldglass/SAP, Accenture) to AI/ML Engineer to Cloud Solutions Architect. Aligned with AI-102 and AWS Solutions Architect Associate prep.

WEEK 1: Python Foundations + Git/GitHub

Day 1: Python setup, variables, data types, strings, f-strings, input()

Day 2: Conditions (if/elif/else), loops (for/while), break/continue, in keyword

Day 3: Functions - def, return values, default parameters, multiple returns, lambda functions

Day 4: Data structures - lists, dictionaries, tuples, sets, slicing, list comprehension, nested dicts

Day 5: File handling - with open(), read/write modes, JSON - json.dump(), json.load()

Day 6: Calling real AI APIs - Groq/Llama 3.3, ask_ai() function, system prompts, conversation memory

Day 7: Intro to RAG - document injection into prompts, hallucination guardrail

Day 8: Git & GitHub - git init, add, commit, branch, merge, push, .gitignore, pushing to GitHub

WEEK 2: SQL + Environment Management

Day 9: SQL basics - what a relational database is, SELECT, WHERE, ORDER BY, LIMIT

Day 10: INSERT, UPDATE, DELETE, primary keys, foreign keys

Day 11: JOIN types - INNER, LEFT, RIGHT, FULL; GROUP BY, HAVING, aggregate functions

Day 12: Subqueries, indexes, basic query optimization

Day 13: Python + SQL - sqlite3 module, running queries from Python

Day 14: Environment management - venv, pip, requirements.txt, pyproject.toml

WEEK 3: RAG Deep Dive

Day 15: Text chunking strategies - fixed-size, sentence-based, recursive splitting, overlap tradeoffs

Day 16: Embeddings - what vectors represent, embedding models, cosine similarity

Day 17: Vector databases - ChromaDB setup, storing/querying embeddings

Day 18: Full RAG pipeline - chunk to embed to store to retrieve to generate

Day 19: Retrieval quality - top-k tuning, re-ranking basics, metadata filtering

Day 20: Project: PDF Chatbot - build end-to-end RAG app over a real PDF

Day 21: Project polish + push to GitHub with README

WEEK 4: LangChain + LangGraph Intro

Day 22: Why LangChain - chains, prompt templates, output parsers

Day 23: LangChain document loaders, text splitters (rebuild RAG using LangChain)

Day 24: LangChain retrievers, memory modules

Day 25: LangChain Expression Language (LCEL) basics

Day 26: Intro to LangGraph - nodes, edges, state graphs (conceptual)

Day 27: Build a simple 2-step LangGraph flow

Day 28: Review + refactor PDF chatbot using LangChain

WEEK 5: Data & ML Foundations

Day 29: NumPy - arrays, indexing, broadcasting, vectorized operations
Day 30: Pandas - Series, DataFrames, reading CSV/Excel, head/info/describe
Day 31: Pandas - filtering, groupby, merging DataFrames, handling missing data
Day 32: Data visualization - Matplotlib basics, Seaborn basics
Day 33: Core ML concepts - supervised vs unsupervised, train/test split, overfitting/underfitting
Day 34: scikit-learn - first model: linear regression
Day 35: scikit-learn - classification: logistic regression, decision trees, evaluation metrics

WEEK 6: FastAPI + Testing + Authentication

Day 36: FastAPI basics - routes, path/query parameters, running with uvicorn
Day 37: Pydantic models - request/response validation, data types
Day 38: Wrap your RAG system as a FastAPI endpoint
Day 39: Error handling - HTTPException, status codes, middleware basics
Day 40: Testing - pytest fundamentals, writing unit tests
Day 41: API testing - TestClient, testing endpoints, mocking external API calls
Day 42: Authentication - API keys, JWT, OAuth2 basics in FastAPI

WEEK 7: PostgreSQL + Redis

Day 43: PostgreSQL setup, relational schema design for your chatbot
Day 44: SQLAlchemy - engine, sessions, models (ORM basics)
Day 45: SQLAlchemy - relationships (one-to-many), migrations with Alembic (intro)
Day 46: Replace JSON file storage with PostgreSQL in your chatbot
Day 47: Redis basics - key-value store, SET/GET, expiry (TTL)
Day 48: Redis for caching AI responses + session storage
Day 49: Integrate Redis caching into your FastAPI chatbot

WEEK 8: Docker + CI/CD Basics

Day 50: Docker concepts - images vs containers, Dockerfile basics
Day 51: Dockerize your FastAPI + PostgreSQL + Redis chatbot
Day 52: Docker Compose - multi-container setup
Day 53: Environment variables, secrets management
Day 54: Intro to CI/CD - what pipelines do, GitHub Actions basics
Day 55: Write a simple GitHub Actions workflow (run tests on push)
Day 56: Project: Dockerized, tested, authenticated RAG API with PostgreSQL + Redis

WEEK 9: Azure Cloud Integration - Part 1 (AI-102 aligned)

Day 57: Azure AI Foundry overview, Azure OpenAI Service setup
Day 58: Connect Python code to Azure-hosted models
Day 59: Azure AI Search - indexing documents, vector search setup
Day 60: Replace ChromaDB with Azure AI Search in your RAG pipeline
Day 61: Azure AI Studio / Prompt Flow - building flows visually
Day 62: Azure AI Content Safety - content filtering basics

Day 63: Review - map everything learned to AI-102 exam domains

WEEK 10: Azure Cloud Integration - Part 2 (Deployment)

Day 64: Azure Document Intelligence - extracting structured data from documents
Day 65: Azure App Service - deployment basics
Day 66: Deploy your Dockerized FastAPI app to Azure App Service
Day 67: Azure Key Vault - secrets management in the cloud
Day 68: Azure monitoring - Application Insights basics
Day 69: Project: Azure-deployed chatbot - live URL, real cloud infra
Day 70: AI-102 practice questions + gap review

WEEK 11: AWS Basics (SAA-aligned)

Day 71: AWS core services review - IAM, EC2, S3 (tied to SAA prep)
Day 72: S3 - storing documents for RAG pipelines
Day 73: AWS Bedrock - overview, available foundation models
Day 74: Call Bedrock models from Python
Day 75: AWS Lambda basics - serverless functions
Day 76: Deploy a simple function to Lambda
Day 77: AWS SAA practice questions + gap review

WEEK 12: Agents

Day 78: What agents are - reasoning + tool use, ReAct pattern (concept)
Day 79: Function calling / tool calling with LLMs
Day 80: Build a custom tool (e.g., calculator function) an LLM can call
Day 81: Build a search tool + combine multiple tools
Day 82: Agent memory and state management
Day 83: Project: AI Travel Planner Agent - multi-tool agent
Day 84: Project polish + push to GitHub

WEEK 13: Multi-Agent Systems + Evaluation

Day 85: LangGraph - multi-step state graphs, conditional edges
Day 86: Multi-agent patterns - supervisor/worker pattern
Day 87: Build a 2-agent system (e.g., researcher + writer)
Day 88: Evaluation - how to test if an AI app actually works, eval metrics
Day 89: Guardrails - input validation, output filtering, preventing prompt injection
Day 90: Project: Multi-agent application
Day 91: Project polish + push to GitHub

WEEK 14: Frontend Basics for Demos

Day 92: HTML basics - structure, forms, elements
Day 93: CSS basics - styling, flexbox/grid for layout
Day 94: JavaScript basics - DOM manipulation, fetch API

Day 95: Streamlit - fastest way to build a demo UI for AI apps

Day 96: Build a Streamlit frontend for your RAG chatbot

Day 97: (Optional) React basics - components, props, state

Day 98: Connect a simple frontend to one of your deployed backends

WEEK 15: MLOps

Day 99: Logging - structured logging in Python apps

Day 100: Monitoring AI apps - tracking latency, errors, token usage

Day 101: Cost tracking - estimating and monitoring LLM API costs

Day 102: Prompt versioning - managing prompt changes over time

Day 103: CI/CD for ML apps - automated testing + deployment pipeline

Day 104: Model/response drift - what it is, why it matters

Day 105: Review week - consolidate MLOps concepts

WEEK 16: Capstone Project

Day 106: Capstone planning - define scope (production AI assistant)

Day 107: Build - backend (FastAPI + PostgreSQL + Redis + auth)

Day 108: Build - AI layer (RAG + agent tools + LangGraph orchestration)

Day 109: Build - frontend (Streamlit or basic React UI)

Day 110: Deploy - Docker + cloud deployment (Azure or AWS)

Day 111: Testing, monitoring, documentation, polished README

Day 112: Final Project: Complete Production AI Assistant - resume centerpiece

Skills Checklist by End of Roadmap

Python, Git & GitHub, SQL & PostgreSQL, APIs & JSON, RAG (chunking, embeddings, vector DBs), LangChain & LangGraph, NumPy, Pandas, ML basics (scikit-learn), FastAPI, Testing (pytest), Authentication (JWT, OAuth, API keys), Redis, Docker & Docker Compose, CI/CD (GitHub Actions), Azure AI (OpenAI Service, AI Search, Prompt Flow, Content Safety, Document Intelligence), AWS (S3, Bedrock, Lambda), AI Agents & multi-agent systems, Evaluation & guardrails, Frontend basics (HTML/CSS/JS/Streamlit), MLOps (logging, monitoring, cost tracking, prompt versioning).

Portfolio Projects

1. PDF Chatbot (Week 3)
2. Dockerized RAG API with DB + Redis (Week 8)
3. Azure-deployed chatbot (Week 10)
4. AI Travel Planner Agent (Week 12)
5. Multi-agent application (Week 13)
6. Complete Production AI Assistant (Week 16, capstone)

Target Roles

AI Engineer (entry-level), GenAI Engineer, Backend Python Developer, Cloud AI Developer, Azure AI Engineer, AI Solutions Developer — long-term: Cloud Solutions Architect